Interaction and Heterogeneous Treatment Effects

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- The ATE, ATT, and ATC are all expectations of the same unit-level contrast, `y(1) - y(0)`, taken over different covariate distributions

- That distinction becomes meaningful only when the individual treatment effect varies across units

Two claims from earlier modules

- When standard methods succeed: the effect was homogeneous, so marginal and conditional effects agree
- In g-computation: without an exposure-covariate interaction, the ATE, ATT, and ATC coincide

This module is where both claims stop holding

What makes the estimands differ

Any difference between them is driven by two things

- Variation in treatment effects across covariate profiles

- Differences in the distribution of those profiles across treated and untreated units

Interaction terms parameterize heterogeneity

A constant treatment effect

- A model like `y ~ a + l` says the conditional mean is `b0 + b1 * a + b2 * l`
- The contrast between `a = 1` and `a = 0` is `b1`, whatever the value of `l`

The treatment effect is assumed to be the same for every unit, regardless of covariate values

Letting the effect vary

- Suppose one variable in the confounder set modifies the treatment effect, and call it `z`
- `y ~ a * z + l` adds the product term: `b0 + b1 * a + b2 * z + b3 * a * z + b4 * l`

The conditional treatment effect becomes `b1 + b3 * z`, which depends on `z`

When is the effect constant?

- If b3 is zero, the treatment effect is constant
- Otherwise the treatment effect changes across values of z

The interaction term formalizes the claim that the effect of treatment differs across levels of a covariate

Interaction or separate models?

- An equivalent representation is to fit separate models within strata of z, which lets the treatment coefficient differ across groups

- The interaction model is often preferred because it estimates those subgroup-specific effects within a single framework, using all the data

A simulation where we know the answer

Simulate the potential outcomes

```r
set.seed(1000)
n <- 10000

sim_data <- tibble(
  l = rnorm(n),
  z = factor(rbinom(n, 1, 0.5), labels = c("low", "high")),
  y0 = 10 + 2 * l + 3 * (z == "high") + rnorm(n),
  y1 = y0 + 2 + 4 * (z == "high"),
  a = rbinom(n, 1, plogis(-1.25 + 0.5 * l + 2.5 * (z == "high"))),
  y = if_else(a == 1, y1, y0)
)
```

What the simulation says

- `l` is a continuous confounder: it affects both the exposure and the outcome

- `z` is a confounder too, and it also modifies the effect: the individual effect is 2 in the `low` stratum and 6 in the `high` stratum

Both potential outcomes are simulated, so the answer is exact in this sample

Who gets treated

```r
sim_data |>
  count(a, z) |>
  mutate(prop = n / sum(n), .by = a)
```

```
# A tibble: 4 × 4
      a z         n  prop
  <int> <fct> <int> <dbl>
1     0 low    3815 0.764
2     0 high   1176 0.236
3     1 low    1177 0.235
4     1 high   3832 0.765
```

About three quarters of the treated are `high`, and about three quarters of the untreated are `low`

The truth in this simulation

```r
sim_data |>
  mutate(effect = y1 - y0) |>
  summarize(
    ate = mean(effect),
    att = mean(effect[a == 1]),
    atc = mean(effect[a == 0])
  )
```

```
# A tibble: 1 × 3
    ate   att   atc
  <dbl> <dbl> <dbl>
1  4.00  5.06  2.94
```

The ATT is larger than the ATC because the treated are mostly `high`, where the effect is 6

G-computation with an interaction

Fit the interaction model

```r
gcomp_model <- lm(y ~ a * z + l, data = sim_data)

tidy(gcomp_model)
```

```
# A tibble: 5 × 5
  term        estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)     9.99    0.0160     624.        0
2 a               2.04    0.0333      61.2       0
3 zhigh           3.03    0.0330      91.8       0
4 l               1.99    0.0101     196.        0
5 a:zhigh         3.95    0.0465      84.9       0
```

The coefficients are conditional

- `a` is the effect in the `low` stratum, and `a:zhigh` is how much larger the effect is in the `high` stratum

- Together they say the effect is about 2 and about 6

That is the conditional story, and averaging it over a population by eye is another matter

avg_comparisons() for the ATE

```r
avg_comparisons(gcomp_model, variables = "a")
```

```
 Estimate Std. Error   z Pr(>|z|)   S 2.5 % 97.5 %
     4.02     0.0238 169   <0.001 Inf  3.97   4.06

Term: a
Type: response
Comparison: 1 - 0
```

Targeting the ATT and the ATC

```r
# ATT: average over the exposed observations
avg_comparisons(
  gcomp_model,
  variables = "a",
  newdata = filter(sim_data, a == 1)
)

# ATC: average over the unexposed observations
avg_comparisons(
  gcomp_model,
  variables = "a",
  newdata = filter(sim_data, a == 0)
)
```

Same model, three populations

```
# A tibble: 3 × 4
  estimand estimate conf.low conf.high
  <chr>       <dbl>    <dbl>     <dbl>
1 ATE          4.02     3.97      4.06
2 ATT          5.06     5.01      5.11
3 ATC          2.97     2.92      3.02
```

Each estimate matches its simulated truth

Averaging one response surface

Once interaction is introduced, these estimands can differ because they average the same heterogeneous response surface over different covariate distributions

Effects within levels of z

```
avg_comparisons(gcomp_model, variables = "a", by = "z")
```

```
    z Estimate Std. Error     z Pr(>|z|)   S 2.5 % 97.5 %
 low      2.04     0.0333  61.2   <0.001 Inf  1.97   2.10
 high     5.99     0.0333 180.0   <0.001 Inf  5.92   6.05

Term: a
Type: response
Comparison: 1 - 0
```

Testing the difference

```
avg_comparisons(
  gcomp_model,
  variables = "a",
  by = "z",
  hypothesis = ~pairwise
)
```

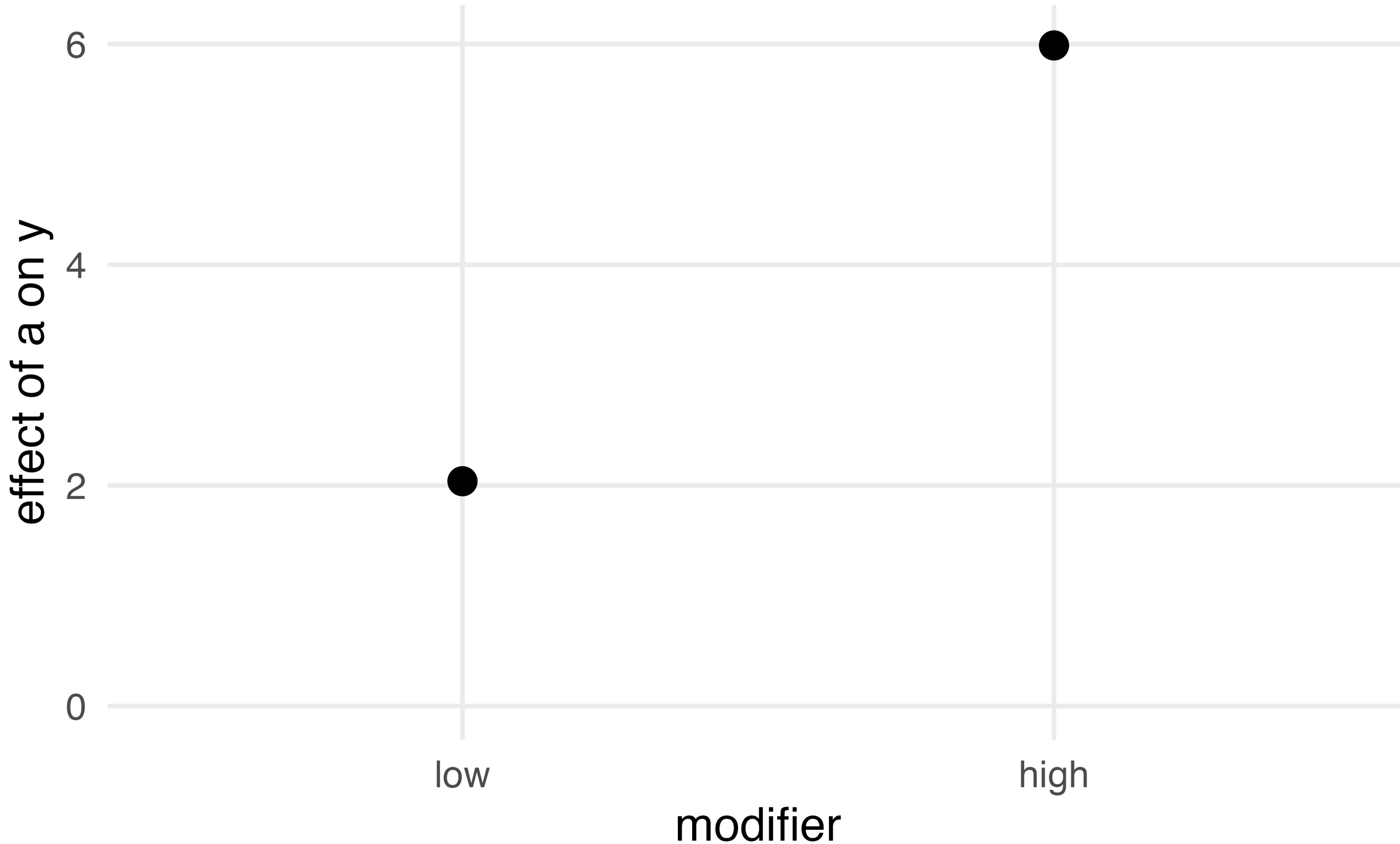
```
      Hypothesis Estimate Std. Error    z Pr(>|z|)   S 2.5 % 97.5 %
 (high) - (low)      3.95     0.0465 84.9   <0.001 Inf  3.86   4.04

Type: response
```

The two strata differ by about 4, with a standard error of about 0.05

The effect in each stratum

```r
plot_comparisons(gcomp_model, variables = "a", by = "z") +
  expand_limits(y = 0) +
  labs(x = "modifier", y = "effect of a on y")
```

The effect in each stratum



Specify every interaction

Because g-computation focuses on modeling the ***outcome***, we must specify every interaction we think exists in order to estimate treatment effect heterogeneity

Drop the interaction

```
gcomp_model_add <- lm(y ~ a + z + l, data = sim_data)
```

```
# A tibble: 3 × 4
  estimand estimate conf.low conf.high
  <chr>       <dbl>    <dbl>     <dbl>
1 ATE          4.01     3.95      4.08
2 ATT          4.01     3.95      4.08
3 ATC          4.01     3.95      4.08
```

Without an exposure-covariate interaction, the three coincide: the model assumes a constant effect across covariate values

Your Turn 1 (`14-interaction-exercises.qmd`)

Fit an outcome model for posted wait times that lets the effect of an Extra Magic Morning vary by ticket season

Use `avg_comparisons()` to get the ATE, then the ATT and the ATC with `newdata`

***Stretch*: refit without the interaction and watch the three collapse**

The same estimands through weights

IPW targets the same estimands

- IPW does not require an explicit regression model for the outcome surface

- The weighting scheme determines which covariate distribution the heterogeneous unit-level effects are averaged over

IPW captures heterogeneity implicitly through the weighting and averaging target, not by estimating subgroup-specific effects

Fit the propensity score model

```
propensity_model <- glm(a ~ l + z, family = binomial(), data = sim_data)

sim_data_wts <- propensity_model |>
  augment(type.predict = "response", data = sim_data) |>
  mutate(
    w_ate = wt_ate(.fitted, a),
    w_att = wt_att(.fitted, a),
    w_atu = wt_atu(.fitted, a)
  )
```

The modifier is a confounder, so it belongs in the propensity score model

Three weighted outcome models

```r
outcome_ate <- lm(y ~ a, data = sim_data_wts, weights = w_ate)
outcome_att <- lm(y ~ a, data = sim_data_wts, weights = w_att)
outcome_atu <- lm(y ~ a, data = sim_data_wts, weights = w_atu)
```

- Each model regresses the outcome on the exposure alone, and only the weights change
- `wt_atu()` builds the weights for the unexposed observations, the ATU or ATC

Estimates through `ipw()`

```r
ipw(propensity_model, outcome_ate) |>
  tidy(conf.int = TRUE) |>
  select(term, estimate, std.error, conf.low, conf.high)
```

```
# A tibble: 1 × 5
  term  estimate std.error conf.low conf.high
  <chr>    <dbl>     <dbl>    <dbl>     <dbl>
1 diff      4.04    0.0404     3.96      4.11
```

The three weighted estimates

```
# A tibble: 3 × 4
  estimand estimate conf.low conf.high
  <chr>       <dbl>    <dbl>     <dbl>
1 ATE          4.04     3.96      4.11
2 ATT          5.03     4.92      5.14
3 ATC          3.03     2.91      3.15
```

The weighted estimates recover the same truths as g-computation

The weights carry the estimand

- The weighted outcome model has no covariates, so the differences among the three estimates flow entirely through the weights

- The modifier separates the exposure groups strongly here, so the spread between the ATT and the ATC is visible

Effect modification with `ipw()`

The subgroup question with weights

```r
outcome_model_int <- lm(y ~ a * z, data = sim_data_wts, weights = w_ate)

ipw(propensity_model, outcome_model_int, .by = z) |>
  tidy(conf.int = TRUE) |>
  select(group, estimate, std.error, conf.low, conf.high)
```

```
# A tibble: 4 × 5
  group                  estimate std.error conf.low conf.high
  <chr>                     <dbl>     <dbl>    <dbl>     <dbl>
1 overall                    4.03    0.0366     3.95      4.10
2 z = low                    2.10    0.0639     1.98      2.23
3 z = high                   5.94    0.0672     5.81      6.07
4 z = high vs z = low        3.84    0.116      3.61      4.06
```

Your Turn 2

Ask whether the effect of an Extra Magic Morning is modified by calendar quarter

Fit the propensity score model, build ATE weights, and pass a weighted outcome model with the interaction to `ipw()` with `.by`

Stabilizing the weights

Stabilized weights, revisited

- `stabilize = TRUE` puts the marginal probability of the observed exposure in the numerator, as it did when we first built ATE weights

- Under effect modification, the numerator should condition on the modifier instead

Condition the numerator on z

```r
numerator_model <- glm(a ~ z, family = binomial(), data = sim_data)
p_z <- fitted(numerator_model)

sim_data_wts <- sim_data_wts |>
  mutate(
    sw_ate = wt_ate(
      .fitted,
      a,
      stabilize = TRUE,
      stabilization_score = if_else(a == 1, p_z, 1 - p_z)
    )
  )
```

`fitted()` alone gives the probability of exposure for every unit, which is not the probability of the exposure an unexposed unit took

The rule for the numerator

- Conditioning the numerator on a variable that is also in the outcome model does not change the estimand

- Conditioning on a variable that is not in the outcome model does change it: the pseudo-population is one where that variable still predicts exposure

Only condition the numerator on variables that are already in the outcome model

The weights tighten

```r
sim_data_wts |>
  select(w_ate, sw_ate) |>
  # the weights stack only as plain numbers
  mutate(across(everything(), as.numeric)) |>
  pivot_longer(everything(), names_to = "weights") |>
  summarize(sd = sd(value), max = max(value), .by = weights)
```

```
# A tibble: 2 × 3
  weights     sd   max
  <chr>    <dbl> <dbl>
1 w_ate    1.50  15.0
2 sw_ate   0.221  3.53
```

Conditioning the numerator on z cuts the spread of the weights and pulls in the largest ones

The same fit, stabilized

```r
outcome_model_stab <- lm(y ~ a * z, data = sim_data_wts, weights = sw_ate)

ipw(propensity_model, outcome_model_stab, .by = z) |>
  tidy(conf.int = TRUE) |>
  select(group, estimate, std.error, conf.low, conf.high)
```

```
# A tibble: 4 × 5
  group                   estimate std.error conf.low conf.high
  <chr>                      <dbl>     <dbl>    <dbl>     <dbl>
1 overall                     4.03    0.0366     3.95      4.10
2 z = low                     2.10    0.0639     1.98      2.23
3 z = high                    5.94    0.0672     5.81      6.07
4 z = high vs z = low         3.84    0.116      3.61      4.06
```

Three jobs for interaction terms

The same syntax, three jobs

- **Functional form**: an interaction between two covariates, in either model
- **Effect modification**: an interaction between the exposure and a covariate
- **Causal interaction**: an interaction between two exposures

Effect modification versus causal interaction

Two different causal questions

- **Effect modification** asks whether the exposure's effect differs across levels of a pre-treatment variable

- **Causal interaction** asks whether the joint effect of changing both variables differs from what their separate effects predict

These are often used interchangeably, but they answer different questions

Effect modification

- The comparison is `E[y(1) - y(0) | z = high]` against `E[y(1) - y(0) | z = low]`

- `z` is used to define subgroups: we ask how the effect of the exposure changes across observed strata of `z`, not what would happen if we intervened on it

Causal interaction

- Both variables are exposures of causal interest
- The comparison is `E[y(a = 1, b = 1) - y(a = 0, b = 1)]` against `E[y(a = 1, b = 0) - y(a = 0, b = 0)]`

The effect of one exposure under two settings of the other

Extra Magic Mornings by quarter

- Does an Extra Magic Morning change posted wait times differently in the fourth quarter than in the first?

- No one can assign a day to a calendar quarter, so the quarter defines subgroups rather than an intervention

That is an effect modification question

Extra Magic Mornings and season

- Disney sets ticket season through pricing, so both the morning policy and the season are things the park can change
- Asking what would happen if we set both at once is a joint intervention

That is a causal interaction question

Intervening on two exposures at once

Two exposures, one confounder

```r
set.seed(1)
n <- 10000

joint_data <- tibble(
  l = rnorm(n),
  a = rbinom(n, 1, plogis(-0.5 + 0.8 * l)),
  b = rbinom(n, 1, plogis(-0.3 + 0.9 * l)),
  y = 10 + 2 * l + 1 * a + 2 * b + 2 * a * b + rnorm(n)
)
```

`l` confounds both exposures, which are otherwise independent given `l`

What this simulation says

- The four counterfactual means are 10, 11, 12, and 15
- The effect of a is 1 when b is 0 and 3 when b is 1
- The difference between those two effects, the interaction, is 2

Cross the two exposures

```r
joint_data <- joint_data |>
  mutate(joint = joint_exposure(a = a, b = b))

levels(joint_data$joint)
```

```
[1] "a = 0, b = 0" "a = 1, b = 0" "a = 0, b = 1" "a = 1, b = 1"
```

The crossing is one categorical exposure with a cell per combination

One model over the cells

```
ps_joint <- multinom(joint ~ 1, data = joint_data, trace = FALSE)

joint_data <- joint_data |>
  mutate(w_joint = wt_ate(fitted(ps_joint), joint))
```

The categorical route takes the matrix of fitted probabilities, one column per cell

The weighted outcome model

```r
joint_model <- lm(y ~ joint, data = joint_data, weights = w_joint)

ipw(ps_joint, joint_model) |>
  tidy(conf.int = TRUE) |>
  select(contrast, group, estimate, conf.low, conf.high)
```

The weighted outcome model

```
# A tibble: 9 × 5
  contrast       group             estimate conf.low conf.high
  <chr>          <chr>                <dbl>    <dbl>     <dbl>
1 a = 0, b = 0 overall              10.0      9.92      10.1
2 a = 1, b = 0 overall              11.0     10.9       11.1
3 a = 0, b = 1 overall              12.0     11.9       12.1
4 a = 1, b = 1 overall              15.0     14.9       15.2
5 a: 1 vs 0    b = 0                 0.982    0.870      1.09
6 a: 1 vs 0    b = 1                 3.02     2.86       3.18
7 b: 1 vs 0    a = 0                 1.98     1.86       2.09
8 b: 1 vs 0    a = 1                 4.01     3.85       4.17
9 a: 1 vs 0    b = 1 vs b = 0        2.04     1.84       2.24
```

Nine rows for a two-by-two

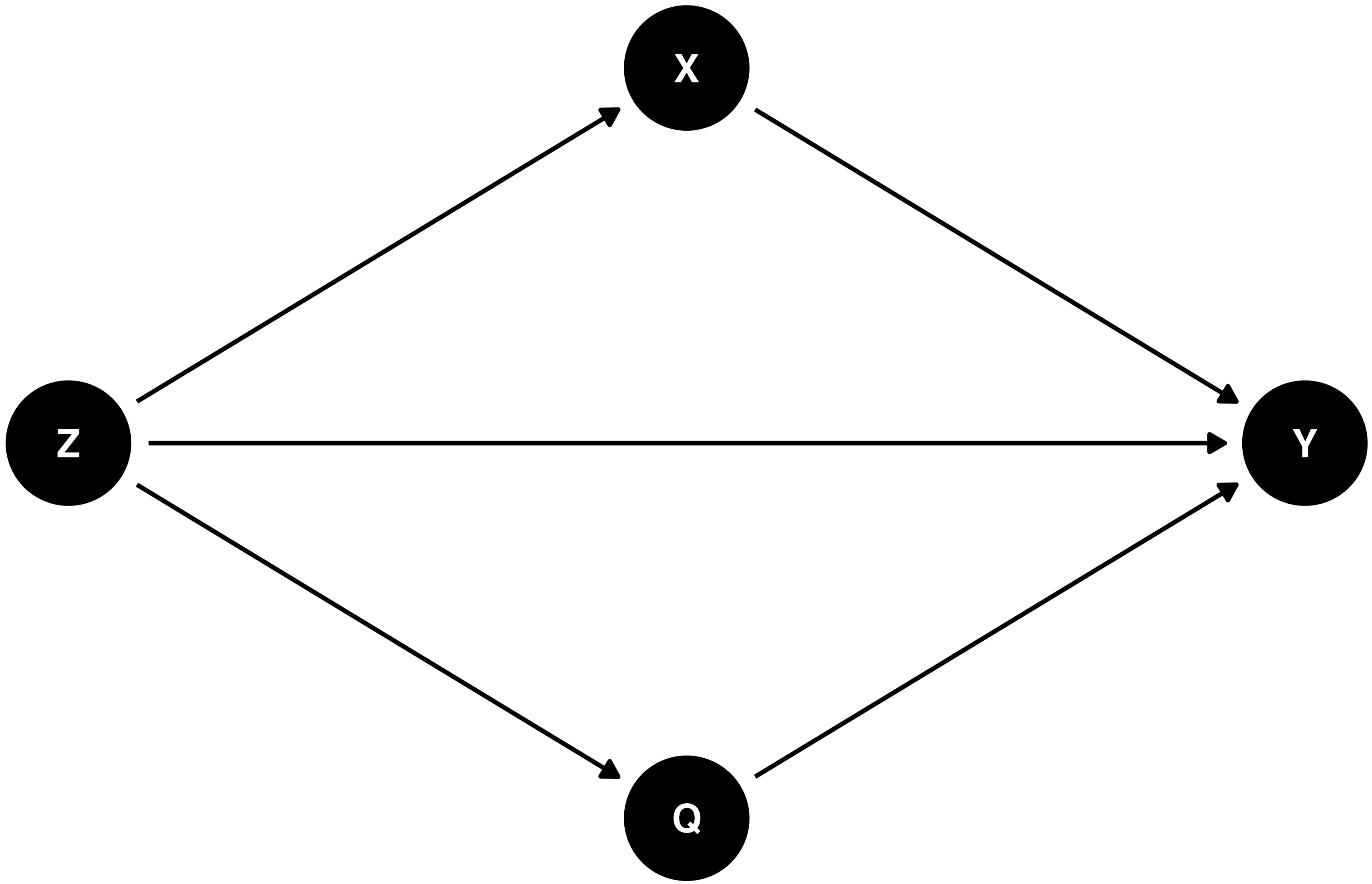
- Four **cell means**: the outcome had everyone been set to that combination
- Four **simple effects**: each exposure's effect within a fixed level of the other
- One **interaction**: the difference between the first exposure's two simple effects

Interaction in a DAG

A conventional DAG

```
conventional_dag <- dagify(
  X ~ Z,
  Q ~ Z,
  Y ~ X + Q + Z,
  coords = time_ordered_coords(),
  exposure = "X",
  outcome = "Y"
)

ggdag(conventional_dag) + theme_dag()
```

A conventional DAG



Interaction is not in the graph

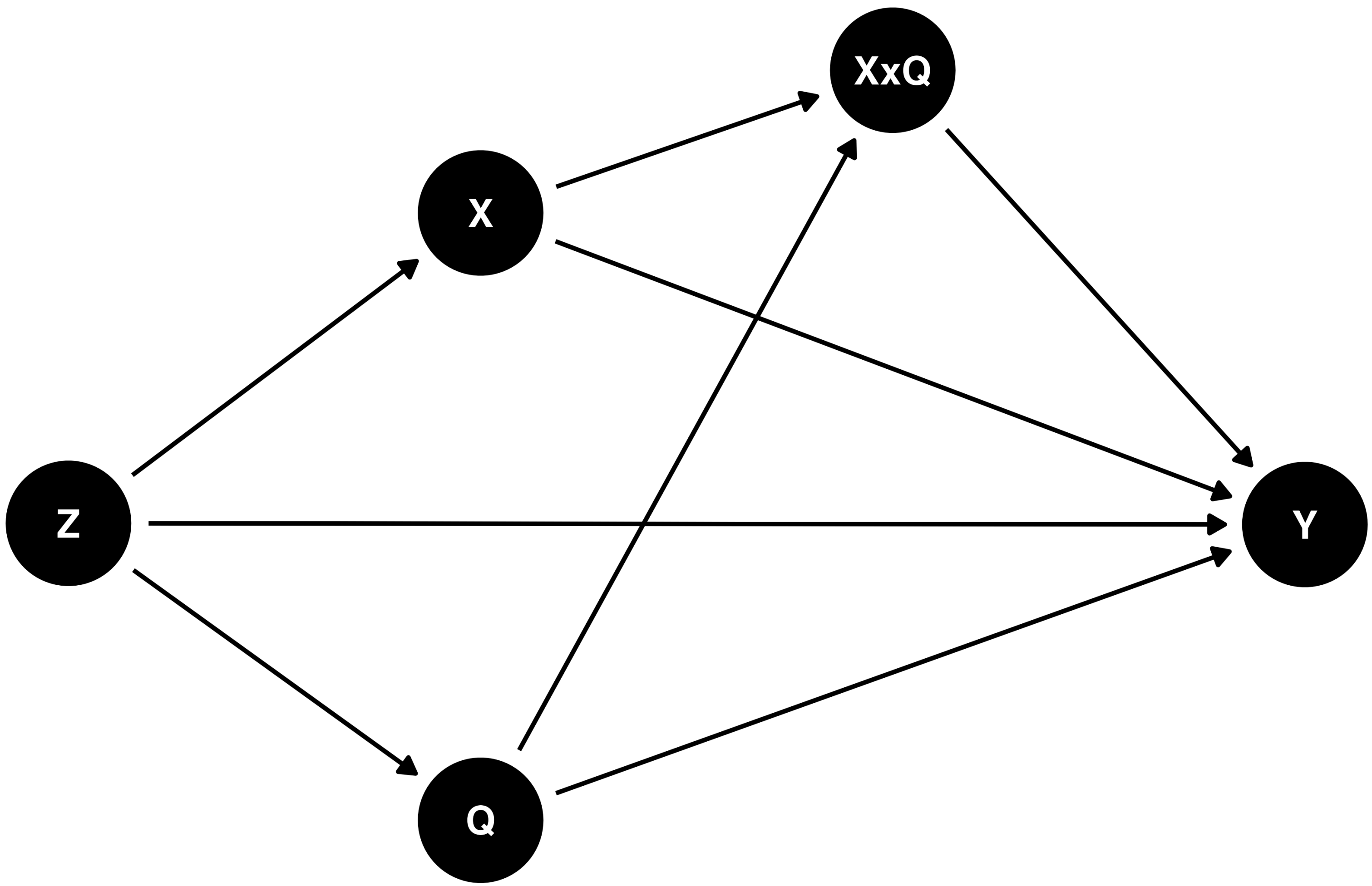
A conventional DAG shows that X, Q, and Z all affect Y, but not whether X and Q interact

The interaction, if present, is implicit in the functional form of how they jointly affect Y

An explicit interaction node

```r
explicit_interaction_dag <- dagify(
  X ~ Z,
  Q ~ Z,
  Y ~ X + Q + Z + XxQ,
  XxQ ~ X + Q,
  coords = time_ordered_coords(),
  exposure = "X",
  outcome = "Y"
)

ggdag(explicit_interaction_dag) + theme_dag()
```

An explicit interaction node



Making the interaction visible

Arrows run from `X` and `Q` into `XxQ`, and from `XxQ` into `Y`, which can help identify adjustment sets for estimating interaction effects

Attia J, Holliday E, Oldmeadow C (2022). A proposal for capturing interaction and effect modification using DAGs. International Journal of Epidemiology. doi:10.1093/ije/dyac126

Your Turn 3

Ask the joint question: what if the park set both the Extra Magic Morning policy and the ticket season?

Cross the two with `joint_exposure()`, fit one propensity model over the cells, and read the cell means, simple effects, and interaction

Finish early? Try the *stretch goal*

10:00